

Vector Calculus

Problem Sheet 5

1. For each of the following functions f , compute ∇f

(a) $f(x, y) = x^2 - x^3y^2 + y^4$, (b) $f(x, y) = y - e^{-2x^2y}$,
(c) $f(x, y, z) = \frac{xy^2}{z^3}$, (d) $f(x, y, z) = xy \cos(yz)$.

2. If

(a) $\phi(x, y, z) = 2x^2y + 3xyz + yz^3$, (b) $\phi(x, y, z) = x^2z + \exp\left(\frac{y}{x}\right)$,

evaluate $\text{grad } \phi$ and $\text{div}(\text{grad } \phi)$ at the point $(1, 2, -1)$.

3. If

$$\mathbf{v} = 2xy\mathbf{i} + z^2\mathbf{j} + xyz\mathbf{k},$$

evaluate $\text{div } \mathbf{v}$ and $\text{curl } \mathbf{v}$.

4. Evaluate the divergence and curl of the following vector fields:

(a) $\mathbf{F}(x, y, z) = xz\mathbf{i} + yz\mathbf{j} + xy\mathbf{k}$,
(b) $\mathbf{F}(x, y, z) = 3x^2y\mathbf{i} + 2xz^3\mathbf{j} + y^4\mathbf{k}$,
(c) $\mathbf{F}(x, y, z) = xe^{-z}\mathbf{i} + 4yz^2\mathbf{j} + 3ye^{-z}\mathbf{k}$,
(d) $\mathbf{F}(x, y, z) = x^2 \sin(yz)\mathbf{i} + z \cos(xz^3)\mathbf{j} + ye^{5xy}\mathbf{k}$.

5. Let $\mathbf{a}$ be a constant vector and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Verify the following identities:

(a) $\text{div } \mathbf{r} = 3$,
(b) $\text{curl } \mathbf{r} = \mathbf{0}$,
(c) $(\mathbf{a} \times \nabla) \times \mathbf{r} = -2\mathbf{a}$,
(d) $\nabla \cdot (\mathbf{a} \times \mathbf{r}) = 0$,
(e) $\nabla \times [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2(\mathbf{r} \times \mathbf{a})$,
(f) $\nabla \cdot [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2(\mathbf{r} \cdot \mathbf{a})$.

6. If

$$\mathbf{A} = (2x^2, -3yz, xz^2) \quad \text{and} \quad \phi = 2z - x^3y$$

evaluate $\mathbf{A} \cdot \nabla \phi$ and $\mathbf{A} \times \nabla \phi$ at the point $(1, -1, 1)$.

7. Find the directional derivative of

$$\phi(x, y) = \tan^{-1} \left(\frac{y}{x} \right),$$

at the point $(2, -2)$ in the direction of the unit vector of $\mathbf{i} - 3\mathbf{j}$.

8. Find the directional derivative of

$$\phi(x, y, z) = xy^2z + 4yz^2$$

at the point $(-2, 1, -1)$ in the direction of the unit vector of $-\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$.

9. (a) A vector field is said to be irrotational if $\nabla \times \mathbf{A} = \mathbf{0}$. Find constants a, b, c such that

$$\mathbf{A} = (2x + y^2 + az)\mathbf{i} + (bxy - 4y - z)\mathbf{j} + (3x + cy + 2z)\mathbf{k}$$

is irrotational.

(b) Given a vector field $\mathbf{A}(x, y, z) = (A_1, A_2, A_3)$,

show that for **any** vector field $\mathbf{A}$,

$$\text{div}(\text{curl} \mathbf{A}) = 0.$$

10. Given that $\mathbf{v}(x, y, z) = (x^2yz, xy \log y, e^x \sin z)$, calculate $\nabla \cdot \mathbf{v}$.

Prerequisites:

$P(n) = \{\text{questions with prerequisite Parts } \leq n\}$

$A = \text{all of PS5.}$

$P(13) = \{1, 2, 7\}$

$P(14) = A$

$P(14^*) = A$ (note that Part 14*, on line integration and Green's Theorem, will not be examinable)